

Submit your Jupyter notebook with the answers to gradescope by 3:40pm.

1. Consider $P = (1, 0)$ as a point on the curve $(x^2 + y^2 + 2y)^2 - (x^2 + y^2) = 0$.
 - (a) Compute the tangent line to the curve at P .
 - (b) What is the slope of the tangent line at P ?
2. Plot the curve defined by $(x^2 + y^2 + 2y)^2 - (x^2 + y^2) = 0$ selecting good ranges for x and y .

Convert into polar coordinates and plot.

Describe the difference between the two plots.